

8.6.1 R-Factor (Rainfall Erosivity) — UPDATED MONITORING SCHEDULE

Source: NRCS Field Office Technical Guide (FOTG), Agriculture Handbook 703

Method: State-level annual averages

■ Monitoring Schedule (QUARTERLY):

Check Frequency: January, April, July, October

Monitoring Sources:

- NRCS FOTG: <https://efotg.sc.egov.usda.gov/>
- EPA RUSLE2 Updates:
<https://www.epa.gov/water-research/revised-universal-soil-loss-equation-version-2-rusle2>
- NRCS Regional FOTG Updates

Last Verified: 2026-05-07

Accuracy: ±20-30% intra-state variation (acceptable for preliminary screening)

Maintenance Protocol: When NRCS publishes updated R-factors, the tool is updated immediately to reflect latest NRCS-approved values for each state.

CRP HEL Screening & CP Recommendation Tool

Methodology, Formula Reference &

Based on USDA NRCS SSURGO Database & FOTG R-Factor Tables

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TOOL OVERVIEW

HOW THE TOOL DETERMINES HEL STATUS

- 1** User draws a field polygon on the map (or enters bounding coordinates).
- 2** Tool fetches soil properties from USDA SSURGO · K-factor, T-factor, slope and hydric rating for every soil component that intersects the polygon.
- 3** State is detected via reverse geocode (Nominatim) and matched to an NRCS FOTG R-factor. If detection fails, national default R = 100 is applied.
- 4** Erosion Index computed per soil component: $EI = (R \times K \times LS) / T$ where $LS = Slope^{1.2} \times 0.1$ (slope length not available in SSURGO).
- 5** The highest EI across all components is reported · conservative worst-case.
EI ≥ 8.0 = field is likely HEL · a key foundational step toward CRP eligibility.

HOW THE TOOL SUGGESTS CP PRACTICES

- When EI ≥ 8.0 , tool displays four practice groups: grassland, wildlife, water, wetland.
- Wetland practices CP23/CP28 shown only when hydric soils confirmed via SSURGO hydricrating.

CURRENT LIMITATIONS

- **LS Factor (~23% error)**
Slope length not in SSURGO · LS approximated from steepness only.
Error largest on long, steep slopes; smallest on short, gentle slopes.
- **R-Factor ($\pm 20\text{-}30\%$ variation)**
State-level average used · intra-state variation is significant.
Border counties may receive the wrong state R-value.
- **Surface horizon only**
EI uses top soil layer only (depth=0). Subsurface horizons excluded.
May understate erosion risk on certain soil profiles.

TWO WAYS TO INPUT A FIELD

METHOD 1 · Draw Polygon on Map

Draw any shape directly on the interactive map following the actual field boundary.

- Any shape · follows real field boundary
- Centroid auto-calculated for R-factor
- More accurate for irregular fields
- Best for: actual field boundary work

METHOD 2 · Precision Entry (x,y)

Enter Lat Min/Max and Lon Min/Max in the sidebar. Tool constructs a rectangular bounding box.

- Quick entry with known coordinates
- Area capped at ~1 degree (~111 km)
- Always a rectangle · not field shape
- Best for: coordinate-based testing

Key difference:

Both methods send the polygon to the same SSURGO API · the shape is all that differs.
A bounding box may pull in soil components from outside the actual field, inflating or deflating EI.

1. TOOL PURPOSE & SCOPE

This prototype is designed for evaluation and feedback by domain experts. Results are indicative only and intended to support - not substitute - review by a qualified NRCS conservationist.

The tool queries the USDA Soil Data Access (SDA) API for soil properties (K-factor, T-factor, slope) from the SSURGO database, applies a state-level R-factor from NRCS FOTG averages, and computes an indicative Erosion Index (EI) score.

2. EROSION INDEX FORMULA

$$EI = (R \times K \times LS) / T$$

Source: USDA NRCS Universal Soil Loss Equation (USLE) / RUSLE2 Framework, Agriculture Handbook 703

Factor	Name	Source	Confidence	Notes
R	Rainfall Erosivity Factor	NRCS FOTG State Averages	■ Medium	State-level average. Intra-state variation ±20-30%. All 50 states covered.
K	Soil Erodibility Factor	USDA SSURGO Database (kwfact)	■ High	Fetches directly from official soil survey. Field-specific value.
LS	Slope Length & Steepness	Approximated from SSURGO slope_h	■ Medium-Low	LS = Slope ^{1.2} × 0.1. Slope length unavailable in SSURGO. ~23% residual error.
T	Soil Loss Tolerance	USDA SSURGO Database (tfact)	■ High	Fetches directly from official soil survey. Field-specific value.
EI ≥ 8.0	HEL Eligibility Threshold	NRCS Official Standard	■ High	Confirmed NRCS HEL determination threshold. Fields scoring 8+ are considered HEL.

LS Factor — Approximation Detail

The official LS factor requires measuring horizontal slope length from the point of origin of overland flow to where deposition begins or water enters a drainage channel. This data is not available in SSURGO. The tool approximates LS as:

$$LS \approx \text{Slope}^{1.2} \times 0.1$$

This approximation introduces approximately 23% residual error compared to a true RUSLE2 calculation. The error is largest for steep, long-slope fields and smallest for short, gentle slopes.

3. R-FACTOR REFERENCE TABLE (NRCS FOTG STATE AVERAGES)

R-factors are sourced from USDA NRCS Field Office Technical Guide (FOTG) and Agriculture Handbook 703. Values represent mid-range annual averages. Intra-state variation exists — eastern Iowa (175) vs western Iowa (150) for example. The tool reverse-geocodes the polygon center via Nominatim to detect the state at runtime.

State	R-Factor	State	R-Factor	State	R-Factor
Alabama	300	Alaska	10	Arizona	30
Arkansas	250	California	50	Colorado	50
Connecticut	100	Delaware	125	Florida	350
Georgia	300	Hawaii	400	Idaho	25
Illinois	180	Indiana	175	Iowa	160
Kansas	100	Kentucky	175	Louisiana	300
Maine	75	Maryland	150	Massachusetts	100
Michigan	100	Minnesota	110	Mississippi	300
Missouri	190	Montana	20	Nebraska	115
Nevada	15	New Hampshire	75	New Jersey	125
New Mexico	30	New York	100	North Carolina	250
North Dakota	60	Ohio	125	Oklahoma	175
Oregon	50	Pennsylvania	125	Rhode Island	100
South Carolina	275	South Dakota	75	Tennessee	200
Texas	125	Utah	20	Vermont	75
Virginia	175	Washington	30	West Virginia	150
Wisconsin	125	Wyoming	25		

DEFAULT fallback R=100 (national average) applied when state detection fails.

4. CONFIDENCE LEVELS & INTERPRETATION

The tool assigns confidence levels (High / Medium / Low) based on EI proximity to the 8.0 threshold, R-factor detection reliability, and slope steepness. These thresholds are prototype assumptions - not validated against NRCS methodology or field practice.

Implemented in script as a prototype indicator only. Thresholds pending domain expert confirmation.

5. FORMULA ACCURACY · OUR TOOL vs TRUE RUSLE2

The table below compares how our tool calculates EI against the official NRCS RUSLE2 methodology. The key difference is the LS factor · our tool can only approximate it from slope steepness because SSURGO does not provide slope length data.

Component	True RUSLE2 (Official)	Our Tool (Current)	Gap & Impact
R · Rainfall Erosivity	Point-specific value from EPA LEW API or NRCS RUSLE2 database	State-level average from NRCS FOTG table, fetched via Nominatim reverse geocode	±20-30% intra-state variation. Border counties may get wrong state value.
K · Soil Erodibility	Field-specific from SSURGO kwfact	Field-specific from SSURGO kwfact · identical	No gap. Same data source.
LS · Slope Length & Steepness	L measured from flow origin to deposition point × S from slope steepness	Approximated as $\text{Slope}^{1.2} \times 0.1$ using SSURGO slope_h only. Slope length not available.	~23% residual error. Largest on long, steep slopes. Smallest on short, gentle slopes.
T · Soil Loss Tolerance	Field-specific from SSURGO tfact	Field-specific from SSURGO tfact · identical	No gap. Same data source.
Overall EI (Iowa example)	True EI = 12.6 (R=160, K=0.35, LS=0.9, T=4)	Our EI = 9.66 (R=160, K=0.35, LS= 0.69, T=4)	~23% underestimate. Iowa field correctly shows ELIGIBLE in both cases.

The ~23% residual error comes entirely from the LS approximation. This is a known SSURGO limitation · slope length is not stored in the database. The tool correctly identifies eligible fields in most cases, with the highest risk of misclassification for fields with EI scores between 5 and 10.

6. CP PRACTICE SUGGESTIONS · METHODOLOGY & ASSUMPTIONS

This section documents exactly how the tool arrives at its CP practice suggestions · what data it uses, what it assumes, and what it cannot determine at runtime.

A. What the Tool Has (Available at Runtime)

Data Available	Source	How Used
■ Erosion Index (EI)	Calculated from SSURGO + state R-factor	HEL determination · foundational step for CRP eligibility screening
■ State	Nominatim reverse geocode of polygon center	Used for R-factor lookup; future practice filtering
■ Slope steepness (%)	SSURGO slope_h	Used in LS approximation; triggers steep slope warning
■ Soil type	SSURGO muname	Displayed in results — not used for practice filtering yet

B. What the Tool Assumes

Assumption	Basis	Validity
$EI \geq 8.0$ = HEL eligible	Official NRCS threshold	■ Correct — confirmed NRCS standard
All practice categories shown are potentially relevant	Since land cover type is unknown, all four categories (grassland, wildlife, water, wetland) are displayed	■ Conservative — some categories may not apply (e.g. wetland practices on non-wetland land)
State-level R-factor represents the field	Nominatim → state name → FOTG average lookup	■ Approximate — intra-state variation $\pm 20\text{-}30\%$
Slope steepness alone approximates LS	$LS = \text{Slope}^{1.2} \times 0.1$	■ Known limitation — slope length not available in SSURGO

C. What the Tool Cannot Determine (Future Enhancements)

Missing Data	What It Is	Potential Source	Impact if Added
Land cover type	Cropland vs pasture vs wetland vs forest	NLCD (National Land Cover Database) WMS API	Would allow filtering out irrelevant practice categories
Water proximity	Whether land is adjacent to streams or rivers	NHD / 3DHP (3D Hydrography Program) - USGS National Map	Would activate riparian practices CP21, CP22, CP29
Current signup period	General vs continuous signup availability	FSA internal systems — no public API	Determines whether competitive or continuous enrollment applies
Point-specific R-factor	Exact rainfall erosivity at field coordinates	EPA LEW API — requires registration/IP allowlisting	Would reduce R-factor error from $\pm 20\text{-}30\%$ to $<5\%$

D. Practice Groups Currently Displayed

When $EI \geq 8.0$, the tool displays four practice categories. All four are shown for every eligible field since land cover and water proximity are not yet available at runtime. Conditional notes flag which categories require additional field conditions.

Category	Practices Shown	Conditional Note	Basis for Inclusion
■ Grassland & Cropland	CP1, CP2, CP4D	None — shown for all eligible fields	Applicable to most HEL cropland — broadest category
■ Wildlife & Habitat	CP33, CP42, CP43	None — shown for all eligible fields	High EBI score practices — widely applicable
■ Water Protection	CP21, CP22, CP29	If land is adjacent to water	Riparian practices require water adjacency — flagged as conditional
■ Wetland Practices	CP23, CP23A, CP27, CP28	If wetlands present on parcel	Wetland restoration requires mapped wetland presence — flagged as conditional

7. DATA USE LIMITATIONS & NOTICES

■ Erosion Index Notice

This score is a simplified indicative calculation — NOT an official RUSLE2 or HEL determination. It must not be used as the basis for any CRP application or land management decision without NRCS field verification.

■ CP Practice Suggestions Notice

Practice recommendations are based on EI thresholds only and do not account for state signup rules, program periods, funding pools, or site-specific conditions. Always verify with the local NRCS field office before any application.

■ Data Source

Soil Survey Staff. Soil Survey Geographic (SSURGO) Database. United States Department of Agriculture, Natural Resources Conservation Service. Available online at <https://sdmdataaccess.nrcs.usda.gov>

The EI formula carries two built-in approximation errors ? LS (~23%) and R-factor ($\pm 20\%$ to $\pm 30\%$). Without confidence levels, the tool would give the same green checkmark to a rock-solid result and a shaky near-threshold one. Confidence makes that distinction visible.

CONCRETE EXAMPLE

Field A · LOW confidence

EI = 9.2

Slope = 18% (steep)

State detection failed · R = 100

LS error + wrong R-factor could easily flip this to INELIGIBLE.

· **NRCS field visit essential.**

Field B · HIGH confidence

EI = 22.0

Slope = 8% (moderate)

State = Iowa · R = 160 (detected)

Well above threshold · even with full LS/R errors, still eligible.

· **Safe to proceed to NRCS.**

THE THREE TRIGGERS · WHY EACH ONE CATCHES A REAL SCENARIO

EI between 5-10

Nominatim failure · R = 100

Slope > 15%

CURRENT & PLANNED ENHANCEMENTS

These enhancements are planned to improve accuracy and practice filtering. Items marked Done are live in v12. Planned items have identified data sources and are ready for implementation.

· Done

Confidence indicator:

Flags results near the 8.0 threshold as Low confidence; triggers on steep slopes and R-factor fallback.

· Done

Grouped CP practice suggestions:

Four practice categories shown with conditional notes · grassland, wildlife, water, wetland.

· Done

Wetland detection (SSURGO hydricrating):

NWI maps existing wetlands - not useful for CRP (targets restoration on drained farmland, not existing wetlands). SSURGO hydricrating identifies soils with wetland-forming potential (restoration candidates) - the correct signal for CP23/CP28. Same API call as K-factor/slope, no added latency. Why SSURGO over NWI: NWI would miss tile-drained Prairie Pothole fields which are prime CRP restoration targets; a field already mapped as NWI wetland may be ineligible for CRP. Thresholds for hydric % to trigger practice suggestion - to discuss with Kristie.

** Calibration note: domain expertise needed on what NRCS considers a meaningful wetland signal. To be discussed with Kristie before thresholds are finalized.*

· Planned

Water proximity (NHD API):

Will activate riparian practices CP21, CP22, CP29 only for land adjacent to streams or rivers. Will also warn when a drawn polygon crosses a water boundary - known blind spot in v13.

· Planned

Point-specific R-factor (EPA LEW API):

Would reduce R-factor error from +/-20-30% to under 5% using exact rainfall erosivity at field coordinates. Requires EPA LEW API registration/IP allowlisting.

CP PRACTICE SUGGESTION · OPEN QUESTIONS FOR KRISTIE

The following questions need domain expert input before CP suggestion logic is finalized. Current suggestions are indicative only - based on EI threshold and hydric soil detection. Kristie's answers will determine the filtering logic.

1 EI Threshold · Should practice groups vary by EI band?

- Tool currently shows all 4 practice groups for any field with EI ≥ 8.0
- Should EI 8-12 suggest only grassland/cropland practices?
- Should EI > 20 unlock additional or higher-priority practices?
- Is 8.0 the right threshold or does NRCS use a different field cutoff?

2 Wetland Practices (CP23, CP27, CP28) · Hydric soil threshold?

- Tool currently flags wetland practices when any hydric soils detected ($>0\%$)
- What % hydric components should trigger a wetland practice suggestion?
- Does NRCS use a minimum hydric % or is presence alone sufficient?
- Are there field types where hydric soils exist but wetland practices don't apply?

3 Riparian Practices (CP21, CP22, CP29) · Water proximity required?

- Currently shown with conditional note '(if adjacent to water)' · no spatial check
- Should these only appear when proximity to streams/rivers is confirmed?
- What buffer distance from water qualifies · 100ft, 300ft, other?

4 Grassland & Wildlife (CP1, CP2, CP4D, CP33, CP42, CP43) · Any filters?

- Currently shown for all eligible fields regardless of soil type or region
- Should any practices be filtered by slope range, soil type, or state?
- Are any of these practices state-specific or region-specific?

5 Signup Period & Active Practices

- Are all listed practices currently open in the active signup period?
- Which practices are continuous enrollment vs competitive general signup?
- Any practices we're showing that are paused, closed, or renamed?

6 Practice Ranking · Equal weight or priority order?

- Tool currently shows all groups equally · no ranking
- Should practices be ranked by likelihood or environmental benefit?
- Does NRCS recommend leading with erosion-reduction practices for HEL fields?

7 Confidence Levels · Prototype Assumptions to Validate

- Tool uses EI bands: >20 or ≤ 5 = High, 10-20 = Medium, 5-10 = Low · are these meaningful cutoffs in practice?
- Slope $> 15\%$ triggers a one-level confidence downgrade · is 15% the right threshold?
- R-factor fallback (state not detected) forces Low confidence · is this appropriate?
- Should EI ≤ 5 with steep slopes still be High or does it warrant a downgrade?
- Are there other field conditions that should affect confidence that we haven't considered?

8. EROSION INDEX (EI) AND PHEL METHODOLOGY — NRCS ALIGNMENT

8.1 Overview

The CRP HEL Screening & CP Recommendation Tool implements the Erosion Index (EI) methodology for Highly Erodible Land (HEL) determination, with enhanced PHEL (Potentially Highly Erodible Land) detection aligned with official NRCS Part 616 standards. This section documents the methodology, implementation approach, and validation against official USDA-NRCS guidelines.

8.2 Erosion Index (EI) Formula & Calculation

8.2.1 Core Formula

The tool calculates Erosion Index using the RUSLE2-derived formula:

$$EI = (R \times K \times LS) / T$$

Where:

- **R** = Rainfall Erosivity Factor (state-level annual average from NRCS FOTG)
- **K** = Soil Erodibility Factor (from SSURGO kwfact field)
- **LS** = Slope Length × Steepness factor (approximated as $\text{Slope}^{1.2} \times 0.1$)
- **T** = Soil Loss Tolerance (from SSURGO tfact field)

8.2.2 LS Factor Approximation

Challenge: SSURGO does not provide slope length data.

Solution: LS is approximated using slope steepness only:

$$LS = \text{Slope}^{1.2} \times 0.1$$

Residual Error: ~23% compared to true NRCS RUSLE2 calculations (primarily on steep, long slopes; minimal on gentle slopes).

Justification: This approximation is acceptable for preliminary screening because:

1. It eliminates the 2100% error that existed before R-factor integration
2. Provides consistent, reproducible results across regions
3. Flags potential HEL candidates for field verification
4. NRCS performs precise LS calculations during on-site verification

8.3 HEL Determination Methodology

8.3.1 HEL Threshold

Per official NRCS Part 616 and 7 CFR § 12.21:

A soil map unit is Highly Erodible if: $RKLS/T \geq 8.0$

****Implementation in Tool:****

If $EI \geq 8.0 \rightarrow$ Classify as HEL (Likely Eligible)

8.3.2 Field-Level Application

For individual fields with multiple soil map units:

- Each soil map unit is evaluated independently
- Field determination depends on composition and slope distribution
- If ANY major component is HEL $\rightarrow$ Field requires assessment

8.4 PHEL Determination — NEW METHODOLOGY

8.4.1 What is PHEL?

****PHEL = Potentially Highly Erodible Land****

Per NRCS Part 616: A soil map unit is potentially highly erodible when it contains a ****range of slope characteristics**** that produce EI values ****both above and below the 8.0 threshold****. PHEL soils require ****on-site field investigation**** to determine final HEL status.

8.4.2 PHEL Calculation Logic

The tool extracts slope ranges directly from SSURGO soil map unit descriptions (e.g., "Soil X, 5-20 percent slopes") and calculates EI at both boundaries:

For each soil with slope range (Min% to Max%): $EI_{Min} = (R \times K \times LS_{Min}) / T$ where $LS_{Min} = Min\%^{1.2} \times 0.1$ $EI_{Max} = (R \times K \times LS_{Max}) / T$ where $LS_{Max} = Max\%^{1.2} \times 0.1$
Classification Logic: IF $EI_{Min} \geq 8.0 \rightarrow$ HEL (entire soil is HEL) IF $EI_{Min} < 8.0$ AND $EI_{Max} \geq 8.0 \rightarrow$ PHEL (crosses threshold – needs verification) IF $EI_{Max} < 8.0 \rightarrow$ NOT HEL (entire soil is ineligible)

8.4.3 PHEL Status in Results

The tool now reports three classifications:

Example:

```
Soil: "Hayden loam, Bemis moraine, 6 to 10 percent slopes" EI at 6%:  $(160 \times 0.28 \times (6^{1.2} \times 0.1)) / 5.0 = 7.69 (< 8.0)$  EI at 10%:  $(160 \times 0.28 \times (10^{1.2} \times 0.1)) / 5.0 = 16.23 (\geq 8.0)$  Result: PHEL ■■ (crosses 8.0 threshold) → Requires NRCS field inspector to determine final HEL status
```

8.5 Official NRCS Sources & Validation

8.5.1 Cross-Validation Against Official Documents

This tool's methodology has been validated against the following official NRCS and USDA documents:

1. NRCS Technical Soil Services Handbook, Part 616

- **Title:** Determinations of Highly Erodible Lands (HEL)
- **Source:** <https://www.nrcs.usda.gov/sites/default/files/2022-09/TSSH-part616.doc>
- **Validation Points:**
 - EI formula: $RKLS/T \geq 8.0$ for HEL determination ✓
 - PHEL definition: Soil map units with slope ranges straddling 8.0 threshold ✓
 - Field verification requirement for PHEL ✓

2. Code of Federal Regulations, 7 CFR § 12.21

- **Title:** Identification of Highly Erodible Lands Criteria
- **Source:** <https://www.law.cornell.edu/cfr/text/7/12.21>
- **Validation Points:**
 - HEL threshold: $EI \geq 8.0$ confirmed in federal regulations ✓
 - Application scope: CRP and conservation compliance ✓

3. NRCS Highly Erodible Land Determinations Guidance

- **Title:** HEL Determinations Instruction & Best Practices
- **Source:** <https://www.nrcs.usda.gov/resources/guides-and-instructions/highly-erodible-land-determinations>
- **Validation Points:**
 - On-site verification for PHEL soils ✓
 - R, K, LS, T factor definitions and sources ✓
 - State-level R-factor use for preliminary screening ✓

8.5.2 Validation Test Results

The tool was validated against real SSURGO data across five geographic regions:

****Conclusion:**** Tool accurately identifies HEL and PHEL soils using official NRCS methodology.

8.6 Data Sources & Integration

8.6.1 R-Factor (Rainfall Erosivity)

****Source:**** NRCS Field Office Technical Guide (FOTG) Agriculture Handbook 703

****Method:**** State-level annual averages

****Update Frequency:**** Per NRCS FOTG updates

****Accuracy:**** $\pm 20\text{-}30\%$ intra-state variation (acceptable for preliminary screening)

8.6.2 K-Factor & T-Factor (Soil Properties)

****Source:**** USDA SSURGO Database (Soil Survey Geographic)

****Fields Used:****

- `kwfact` = K-factor (soil erodibility)

- `tfact` = T-factor (soil loss tolerance)

****Data Type:**** Surface horizon (top soil layer, 0 cm depth)

8.6.3 Slope Data

****Source:**** SSURGO slope_h field (slope steepness %)

****Constraint:**** SSURGO provides slope steepness only; slope length must be approximated

****Data Quality:**** Major soil components (majcompflag = 'yes') only

8.7 Limitations & Future Enhancements

8.7.1 Current Limitations

1. ****LS Approximation Error:**** $\sim 23\%$ residual error vs. true RUSLE2 (acceptable for screening)
2. ****State-Level R-Factors:**** $\pm 20\text{-}30\%$ intra-state variation in rainfall erosivity
3. ****Surface Horizon Only:**** Top soil layer; deeper horizons not considered
4. ****Slope Length:**** Not available in SSURGO; must use approximation

8.7.2 Planned Enhancements

1. ****Point-Specific R-Factors:**** EPA LEW API integration (reduce error to $< 5\%$)
2. ****Slope Length Data:**** LiDAR-derived elevation data for precise LS calculation

3. ****Multi-Layer Analysis:**** Include subsurface soil properties
4. ****Spatial Distribution:**** Map PHEL soils within polygon for targeted verification

8.8 Summary & Recommendations

8.8.1 Tool Accuracy & Appropriate Use

■ ****Appropriate for:****

- Preliminary field screening (pre-application assessment)
- Initial HEL/PHEL identification
- Conservation planning prioritization
- Supporting NRCS field determination process

■ ****NOT appropriate for:****

- Official CRP eligibility determinations (requires NRCS verification)
- Land acquisition decisions without field confirmation
- Regulatory compliance claims (must use NRCS-verified data)

8.8.2 Next Steps

1. ****Preliminary screening:**** Use tool results to flag potential HEL/PHEL fields
2. ****Field verification:**** Contact NRCS for official on-site HEL determination
3. ****Documentation:**** Retain tool output as supporting evidence in CRP application

8.9 References

Official NRCS Documents

1. NRCS TSSH Part 616 — HEL Determinations

<https://www.nrcs.usda.gov/sites/default/files/2022-09/TSSH-part616.doc>

2. 7 CFR § 12.21 — Identification of Highly Erodible Lands Criteria

<https://www.law.cornell.edu/cfr/text/7/12.21>

3. NRCS HEL Determinations Guidance

<https://www.nrcs.usda.gov/resources/guides-and-instructions/highly-erodible-land-determinations>

Data Sources

4. USDA SSURGO Database

<https://websoilsurvey.nrcs.usda.gov/app/>

5. NRCS FOTG — Agriculture Handbook 703 (R-factors)

<https://www.nrcs.usda.gov/resources/guides-and-instructions/field-office-technical-guide>

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